

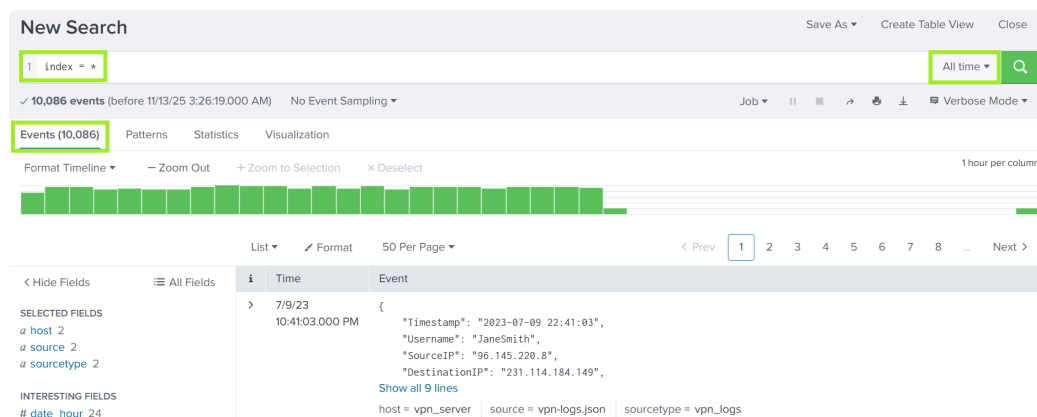
# L2-2.3. Splunk: Dashboards and Reports

## Task 1 - Introduction

Splunk is one of the most widely used Security Information and Event Management (SIEM) solutions in enterprise environments. It helps aggregate data from various data sources within an enterprise environment to enhance security monitoring. However, large volumes of data can quickly overwhelm analysts. In this room, you'll learn practical ways to organize, visualize, and manage data in Splunk to make analysis faster, clearer, and more effective.

## Task 2 - Creating Reports for Recurring Searches

Once Splunk has loaded, head to the **Search & Reporting** app to enter your first query `index = *` and set the time range to `All time` so you can get an overview of the data we will be utilizing in this lab.



**Building Reports:** Heading to the **Reports** tab in Splunk click `Open in Search` to view the query and time settings of the report in the search app.

**Reports**

Reports are based on single searches and can include visualizations, statistics and/or events. Click the name to view the report. Open the report in Pivot or Search to refine the parameters or further explore the data.

7 Reports

| i | Title                           | Actions             | Next Scheduled Time | Owner  | App    | Sharing |
|---|---------------------------------|---------------------|---------------------|--------|--------|---------|
| > | Errors in the last 24 hours     | Open in Search Edit | None                | nobody | search | App     |
| > | Errors in the last hour         | Open in Search Edit | None                | nobody | search | App     |
| > | License Usage Data Cube         | Open in Search Edit | None                | nobody | search | App     |
| > | Messages by minute last 3 hours | Open in Search Edit | None                | nobody | search | App     |
| > | Orphaned scheduled searches     | Open in Search Edit | None                | nobody | search | App     |
| > | Splunk errors last 24 hours     | Open in Search Edit | None                | nobody | search | App     |
| > | Web Connections by Source IP    | Open in Search Edit | None                | admin  | search | Global  |

**Errors in the last 24 hours** Save Save As View Create Table View Close

1 error OR failed OR severe OR sourcetype=access\_\* ( 404 OR 500 OR 503 ) Custom time

Set the time range to **All time**.

```
index = vpn_server | stats count by Username | sort - count
```

This query will search our **vpn\_server** index, give us a count of events based on the **Username** field, and sort them based on the **count** of events. You can see in your Splunk instance and in the screenshot below that we have a total of 86 events, and **Sarah** has the most events recorded. Go ahead and click **Save As** → **Report**.

**New Search** Save As Create Table View Close

1 index = vpn\_server | stats count by Username | sort - count

✓ 86 events (before 11/13/25 4:03:20.000 AM) No Event Sampling

Events (86) Patterns **Statistics (15)** Visualization

100 Per Page Format Preview

| Username | count |
|----------|-------|
| Sarah    | 11    |
| Olivia   | 9     |
| Matthew  | 8     |
| Andrew   | 7     |
| Alice    | 6     |
| Daniel   | 6     |

Report Existing Dashboard New Dashboard Event Type All time

In a SOC environment, you may want to track the users who logged in during a specific time period. Creating a report from the previous query that runs every few hours or daily would be extremely helpful. Let's give our new report a title and description. We used **stats count** in our query, so the **Content** type is automatically

set to **Statistics Table**. You can also choose whether to make the **Time Range Picker** active or inactive in your report. **Save** and **View**.

The left screenshot, titled "Save As Report", shows a form for configuring a report. The "Title" field is "VPN Logins by Username". The "Description" field contains the text "This report provides the number of times each Username logged in via the VPN Server." The "Content" dropdown is set to "Statistics Table". The "Time Range Picker" has two buttons, "Yes" and "No", with "Yes" being the selected option. At the bottom are "Cancel" and "Save" buttons.

The right screenshot, titled "Your Report Has Been Created", shows a confirmation message: "You may now view your report, add it to a dashboard, change additional settings, or continue editing it." Below this, under "Additional Settings:", there is a link for "Permissions". At the bottom are three buttons: "Continue Editing", "Add to Dashboard", and "View".

The screenshot shows the "VPN Logins by Username" report. At the top, it says "This report provides the number of times each Username logged in via the VPN Server." Below this is a green "All time" button. To the right, there are links for "Edit", "More Info", and "Add to Dashboard". Below the description, it shows "✓ 86 events (before 11/13/25 4:03:20.000 AM)". On the right side of the report header, there are icons for "Job", "List", "Refresh", "Share", and "Download". Below the header, it says "15 results" and "20 per page". The main content is a table with two columns: "Username" and "count".

| Username | count |
|----------|-------|
| Sarah    | 11    |
| Olivia   | 9     |
| Matthew  | 8     |
| Andrew   | 7     |

**Scheduling a Report:** Splunk Enterprise unlocks full scheduling capabilities. As shown in the screenshot below, you can configure reports to run on a recurring schedule, such as daily at midnight, using the previous 24 hours of data. Enterprise also allows you to define a report's priority and scheduling window, enabling Splunk to coordinate multiple scheduled reports efficiently and prevent resource usage issues.

## Edit Schedule

×

⚠ Scheduling this report results in removal of the time picker from the report display.

Report VPN Logins by Username

Schedule Report ☒

[Learn More](#)

Schedule Run every day ▾

At 0:00 ▾

Time Range Last 24 hours ▶

Schedule Priority ? Default ▾

Schedule Window ? 5 minutes ▾

Cancel

Save

stats count by Username

| sort -count

This is one of the most commonly used SPL commands for creating quick summaries and identifying top users, hosts, IPs, processes, or other fields.

### Answer the questions below

Head to the Splunk Reports tab and open the Web Connections by Source IP report. Which Source\_IP field value recorded the highest number of events?

10.0.0.1

While viewing the report above, click Edit → Open in Search to investigate the query behind the report. What is the flag value hidden within the search query?

THM{splunk\_report\_wizard!}

The screenshot shows the Splunk Enterprise interface. At the top, there's a navigation bar with 'splunk>enterprise' and 'Apps'. Below it, a menu bar contains 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', and 'Dashboards'. The main heading is 'Web Connections by S...'. Below the heading is a search bar containing the following query:

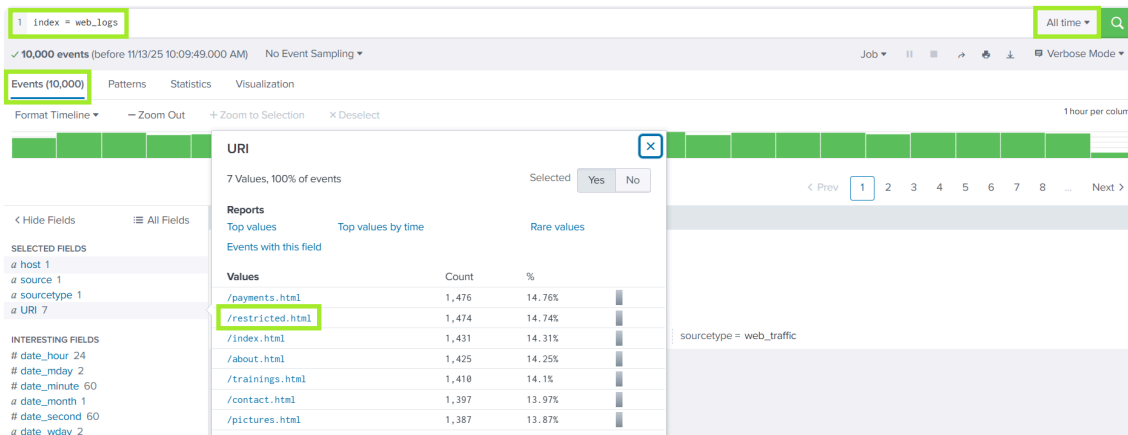
```
1 index = web_logs
2 | stats count by Source_IP
3 | sort - count / THM[splunk_report_wizard!] /
```

Below the search bar, it indicates '✓ 10,000 events (before 03/07/2026 06:33:42.000)' and 'No Event Sampling'. There are four tabs: 'Events (10,000)', 'Patterns', 'Statistics (5)', and 'Visualization'. The 'Statistics (5)' tab is selected. Below the tabs, there are options for '100 Per Page', 'Format', and 'Preview'. The main content area shows a table with the following data:

| Source_IP   |
|-------------|
| 10.0.0.1    |
| 192.0.2.1   |
| 192.168.0.1 |
| 203.0.113.1 |
| 172.16.0.1  |

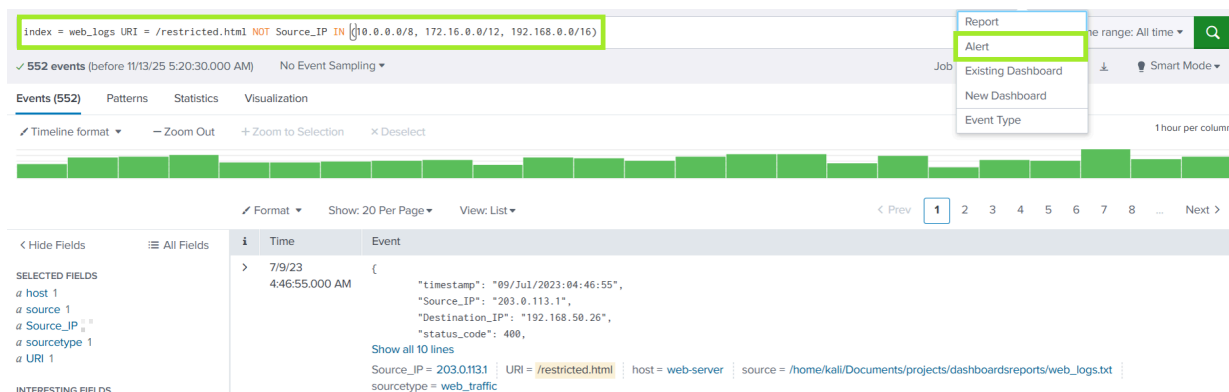
## Task 3 - Detecting With Alerts and Rules

**Building Alerts:** Let's jump into the `web_logs` index and review the available data. We have 10,000 logs available for analysis. Checking out the `URI` field, `/restricted.html` certainly looks interesting.



Let's say that `/restricted.html` should normally only be accessed from internal network IP addresses. To identify potential unauthorized access, we can create a query that returns events where the `Source_IP` falls outside the expected internal ranges. If any external IP addresses appear in the results, it could indicate suspicious activity. After entering the query, we can click `Save As` and `Alert`.

```
index = web_logs URI = /restricted.html NOT Source_IP IN (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16)
```



When saving the alert, you'll need to provide a title and description, and then choose from several configuration options. In the example below, we've selected the following settings:

1. **Alert type:** Real-time - Runs continuously and triggers as soon as a matching event appears

2. **Trigger alert when:** Per-Result - Triggers every time a single event matches the search criteria
3. **When triggered:** Send email - An email will be sent to `soc@tryhackme.com` when the alert is triggered

**Save As Alert**

**Settings**

Title: Restricted URI Accessed by Outside IP

Description: This alert triggers when a `Source_IP` outside of the expected range attempts to access the `/restricted.html` URI.

Permissions: Private | Shared in App

Alert type: Scheduled | **Real-time**

Expires: 24 | hour(s)

**Trigger Conditions**

Trigger alert when: **Per-Result**

Throttle? ☐

**Trigger Actions**

+ Add Actions

When triggered: **Send email**

**When triggered**

Send email

To: `soc@tryhackme.com`

Priority: Normal

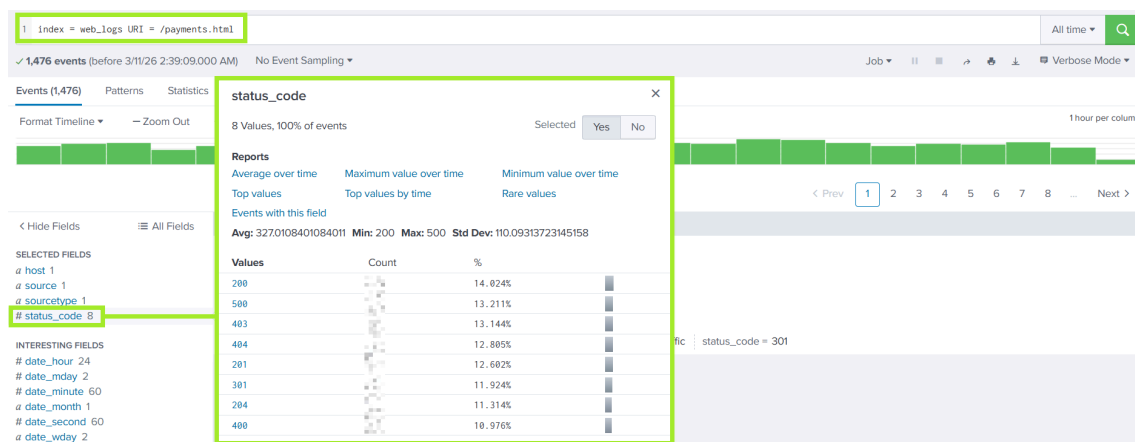
Subject: Splunk Alert: Restricted URI Access

Message: A `Source_IP` outside of the expected range attempted to access `/restricted.html`.

Include: ☒ Link to Alert ☒ Link to Results

**Building a Baseline and Threshold Rule:** Now, let's look at another example using the `web_logs` index again from above. We will specifically focus on the `/payments.html` URI field. If you query this field in your Splunk instance and examine the `status_code` field, you can see that the page returns a range of responses.

index = web\_logs URI = /payments.html

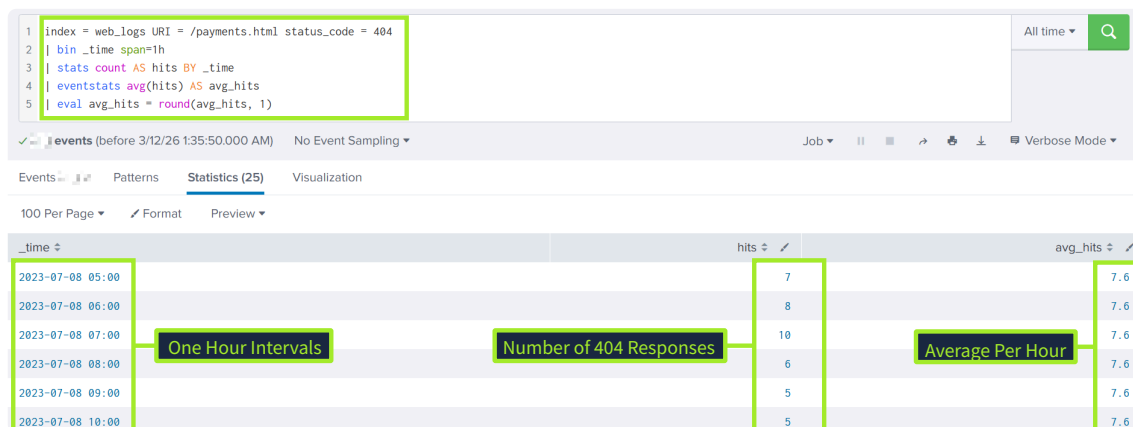


Perhaps we want to build an alert that notifies us if the server has a spike in 404 responses. Well, what do we consider a spike? To answer this, we need to establish a baseline to understand the normal range of this response.

The query below searches the `web_logs` index for `status_code` 404. It first groups events into 1-hour intervals, counts the number of events, and then calculates the average to one decimal, providing a baseline.

```
index = web_logs URI = /payments.html status_code = 404
| bin _time span=1h
| stats count AS hits BY _time
| eventstats avg(hits) AS avg_hits
| eval avg_hits = round(avg_hits, 1)
```

As shown in the screenshot below, `/payments.html` generates an average of 7.6 404 responses per hour based on the available events.



## Threshold

We measured an average of 7.6 404 responses per hour on `/payments.html`. This is our baseline, but traffic isn't static.

- Weekends, holidays, or marketing campaigns can shift the average
- A fixed number like 10 risks false positives, and a larger number could mean missing real issues



Let's create a query that sets a threshold of **>11** in any 1 hour window. This is approximately 1.45 times the average. From this query, you could create an alert that runs every hour and notifies the SOC team of the abnormal response rate.

```
index = web_logs URI = /payments.html status_code = 404
| bin _time span=1h
| stats count AS hits BY _time
| where hits > 11
| eval alert = "HIGH 404s: ".hits." in 1h (normal: ~7.6/hr)"
```

As shown in the screenshot below, if the count exceeds 11, the alert displays the events within the one-hour window, along with the total number of responses received. Just like the previous alert, we can choose to receive a notification in-app or by email, and also configure a variety of additional actions. We can also view the alert's trigger history, allowing us to track when it fired and the results.

The screenshot displays the Splunk Alert configuration and execution interface. It includes a search query, a table of alert results, an alert overview section with configuration details and actions, and a trigger history table.

**Search Query:**

```
1 index = web_logs URI = /payments.html status_code = 404
2 | bin _time span=1h
3 | stats count AS hits BY _time
4 | where hits > 11
5 | eval alert = "HIGH 404s: ".hits." in 1h (normal: ~7.6/hr)"
```

**Alert Results:**

| _time            | hits | alert                                 |
|------------------|------|---------------------------------------|
| 2023-07-08 14:00 | 14   | HIGH 404s: 14 in 1h (normal: ~7.6/hr) |
| 2023-07-08 20:00 | 13   | HIGH 404s: 13 in 1h (normal: ~7.6/hr) |
| 2023-07-08 23:00 | 12   | HIGH 404s: 12 in 1h (normal: ~7.6/hr) |
| 2023-07-09 03:00 | 12   | HIGH 404s: 12 in 1h (normal: ~7.6/hr) |

**404 Response Exceed Threshold**

Enabled: ..... Yes. Disable  
 App: ..... search  
 Permissions: ..... Shared in App. Owned by admin. Edit  
 Modified: ..... Nov 17, 2025 1:33:34 PM  
 Alert Type: ..... Scheduled, Cron Schedule. Edit

Trigger Condition: .. Number of Results is > 0. Edit  
 Actions: ..... 2 Actions Edit  
 Add to Triggered Alerts  
 Send email

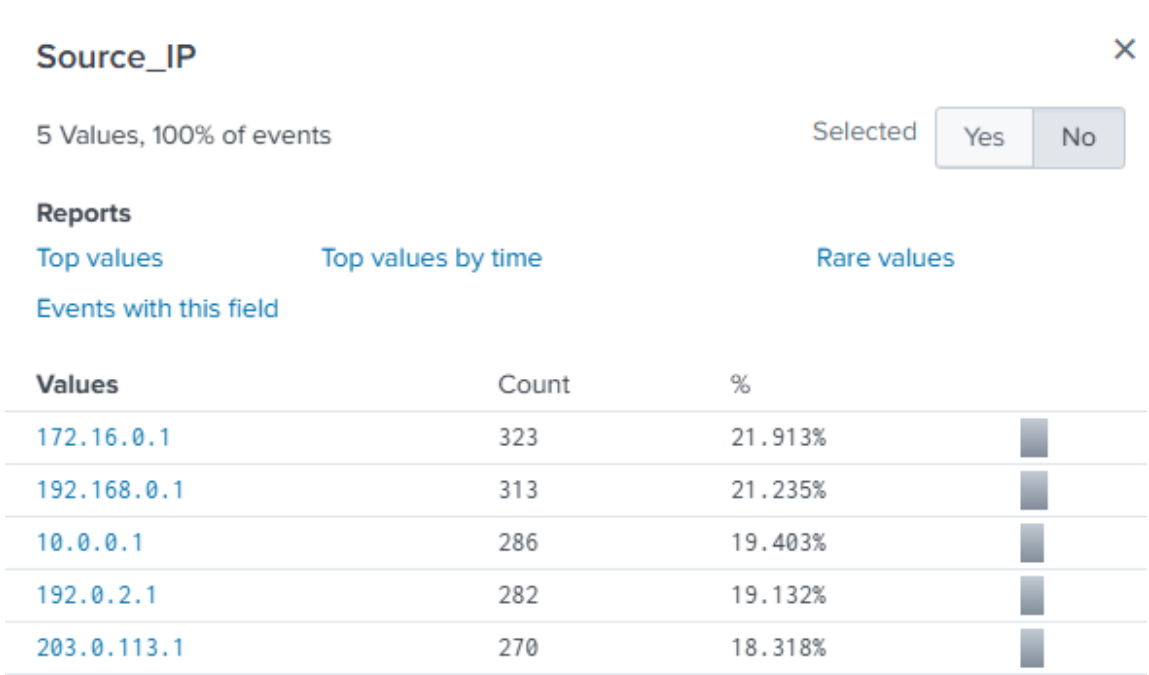
**Trigger History**

|   | TriggerTime              | Actions      |
|---|--------------------------|--------------|
| 1 | 2023-07-09 03:00:00 AWST | View Results |

## Answer the questions below

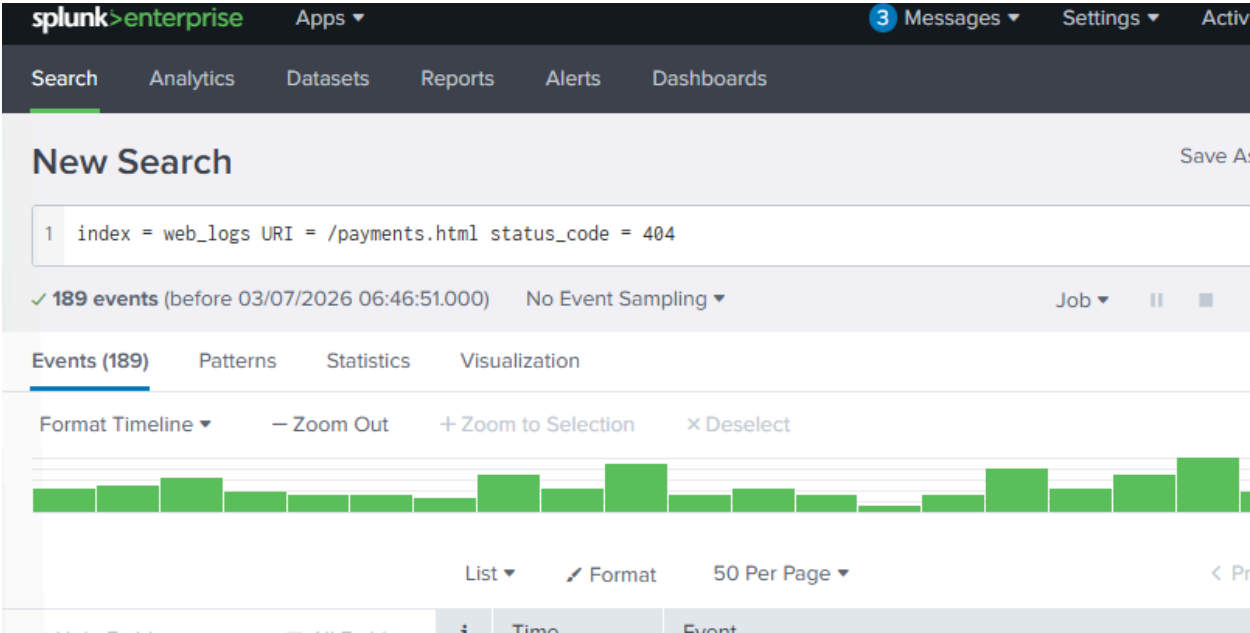
How many Source\_IP addresses outside of the expected range accessed /restricted.html?

2



How many total 404 status\_code were recorded at /payments.html?

189



What is the highest number of 404's received in a given hour with the data available?

16

The screenshot shows the Splunk Enterprise interface. At the top, there's a navigation bar with 'splunk>enterprise' and 'Apps'. Below it, a search bar and tabs for 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', and 'Dashboards'. The 'Search' tab is active, showing a 'New Search' section. The search query is: `1 index = web_logs URI = /payments.html status_code = 404`  
`2 | bin _time span=1h`  
`3 | stats count AS hits BY _time`  
`4 | where hits > 11`  
`5 | eval alert = \"HIGH 404s: \".hits.\" in 1h (normal: ~7.6/hr)\"`  
Below the query, it shows '✓ 189 events (before 03/07/2026 06:49:23.000)' and 'No Event Sampling'. The results are displayed in a table with tabs for 'Events (189)', 'Patterns', 'Statistics (4)', and 'Visualization'. The 'Statistics (4)' tab is selected, showing a table with columns: '\_time', 'hits', and 'alert'. The table contains four rows of data:

| _time            | hits | alert                                 |
|------------------|------|---------------------------------------|
| 2023-07-08 14:00 | 14   | HIGH 404s: 14 in 1h (normal: ~7.6/hr) |
| 2023-07-08 20:00 | 13   | HIGH 404s: 13 in 1h (normal: ~7.6/hr) |
| 2023-07-08 23:00 | 16   | HIGH 404s: 16 in 1h (normal: ~7.6/hr) |
| 2023-07-09 03:00 | 12   | HIGH 404s: 12 in 1h (normal: ~7.6/hr) |

Among these IP addresses:

| IP Address  | Internal (Private) or External (Public) |
|-------------|-----------------------------------------|
| 172.16.0.1  | Internal (Private)                      |
| 192.168.0.1 | Internal (Private)                      |

| IP Address  | Internal (Private) or External (Public) |
|-------------|-----------------------------------------|
| 10.0.0.1    | Internal (Private)                      |
| 192.0.2.1   | External*                               |
| 203.0.113.1 | External*                               |

**Private IP ranges:** These ranges are reserved for internal networks (RFC 1918):

- 10.0.0.0 – 10.255.255.255 ( 10.0.0.0/8 )
- 172.16.0.0 – 172.31.255.255 ( 172.16.0.0/12 )
- 192.168.0.0 – 192.168.255.255 ( 192.168.0.0/16 )

Therefore: 172.16.0.1, 192.168.0.1, and 10.0.0.1 are **internal/private**.

192.0.2.1 and 203.0.113.1 are not private IPs. They belong to special documentation/test ranges (TEST-NET) used in examples and training material, but in Splunk labs and exam questions they are typically treated as **external/non-internal** addresses.

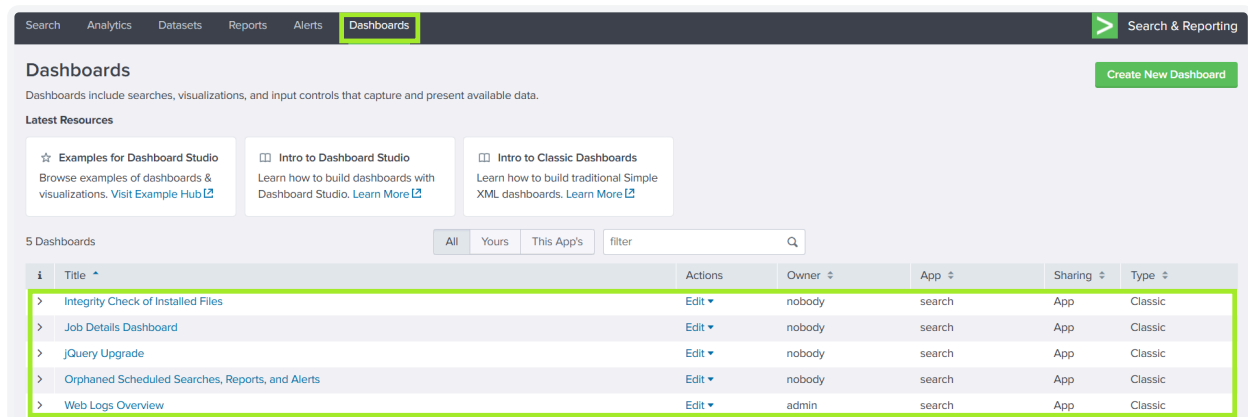
**Answer:** Internal IPs = 172.16.0.1, 192.168.0.1, 10.0.0.1.

External/non-internal IPs = 192.0.2.1, 203.0.113.1.

So there are **2 external IP addresses** in your list.

## Task 4 - Creating Dashboards for Summarizing Results

If you navigate to the **Dashboards** tab in Splunk, you will see a list of default dashboards provided, as well as the one we will be using in this task: **Web Logs Overview**.



Aside from choosing an existing dashboard, you have the option to create a new one in which you must assign a title, provide an optional description, adjust the permissions, and decide whether to use Classic Dashboards or Dashboard Studio.

Dashboard Studio is Splunk's newer dashboard builder, designed to provide users with greater customization options in exchange for a complex learning curve. Classic Dashboards, on the other hand, remain the most commonly encountered format and fully support all standard visualizations. In this room, we will cover Classic Dashboards.

## Create New Dashboard

×

Dashboard Title

Sample Dashboard Title

sample\_dashboard\_title

Edit ID

Description

A Dashboard to Visualize Event Data

Permissions

Private

How do you want to build your dashboard?

What's this?

**Classic Dashboards**  
The traditional Splunk dashboard builder

**Dashboard Studio** NEW  
A new builder to create visually-rich, customizable dashboards

Cancel

Create

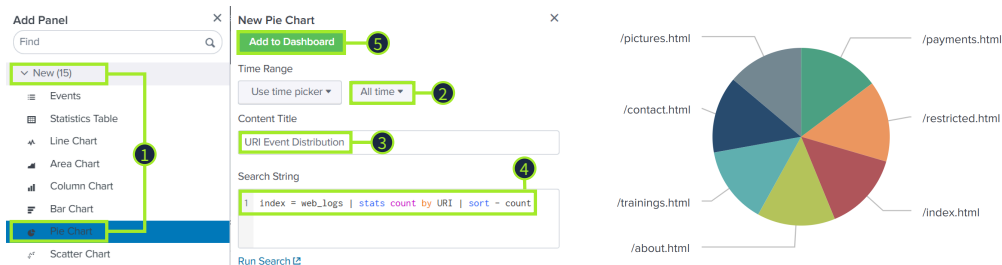
Back in the dashboards tab, go ahead and select the **Web Logs Overview** dashboard. Currently, our dashboard has a single panel that visualizes the count of events over time from the `web_logs` index. This dashboard is helpful, but we can further enhance it by adding more visualizations to better understand

the data from our web server. Go ahead and click **Edit**, then **+ Add Panel** as highlighted in the screenshot below, so we can expand our current dashboard.



Perhaps, we want to build a pie chart that shows the event count for the **URI** field in the **web\_logs** index, helping us visualize how many times each **URI** was accessed within our available events. In the **Add Panel** pop-out, choose:

1. New → Pie Chart
2. Set the time to **All time**
3. Enter a Content Title
4. Enter the search string **index = web\_logs | stats count by URI | sort - count**
5. Click **Add to Dashboard**



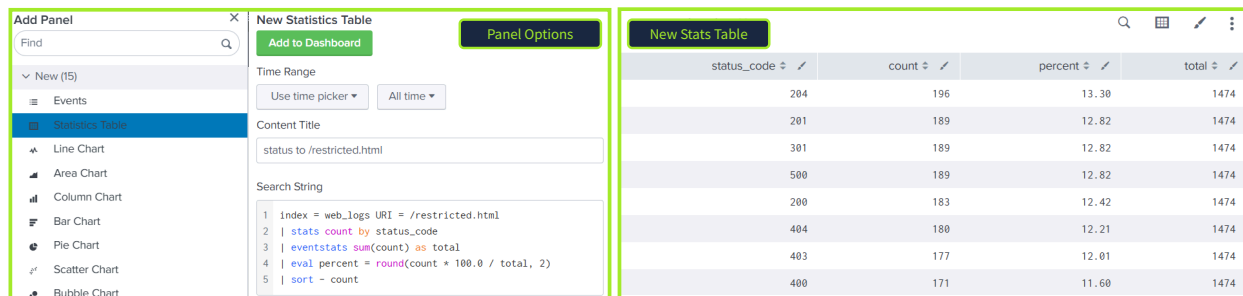
We can add more panels to display any information we like. In the previous task, we looked at the **/restricted.html** **URI** field. Let's create a stats table for our dashboard that shows:

- **status\_code** field

- count of events
- percent of events
- total amount of events overall

Go ahead and click **+ Add Panel** button as you did previously, but this time, choose New → Statistics Table. Next, set your Time Range to All time and enter the following query as the Search String:

```
index = web_logs URI = /restricted.html
| stats count by status_code
| eventstats sum(count) as total
| eval percent = round(count * 100.0 / total, 2)
| sort - count
```



Answer the questions below

Inspect the URI pie chart you built in the dashboard above. Which URI field value has the least amount of events present?

/pictures.html

splunk>enterprise Apps ▾

Search Analytics Datasets Reports Alerts Da

## New Search

```

1 index=*
2 | stats count by URI
3 | sort count

```

✓ 10,086 events (before 03/07/2026 07:11:15.000) No Event Sam

Events (10,086) Patterns **Statistics (7)** Visualization

100 Per Page ▾ ↗ Format Preview ▾

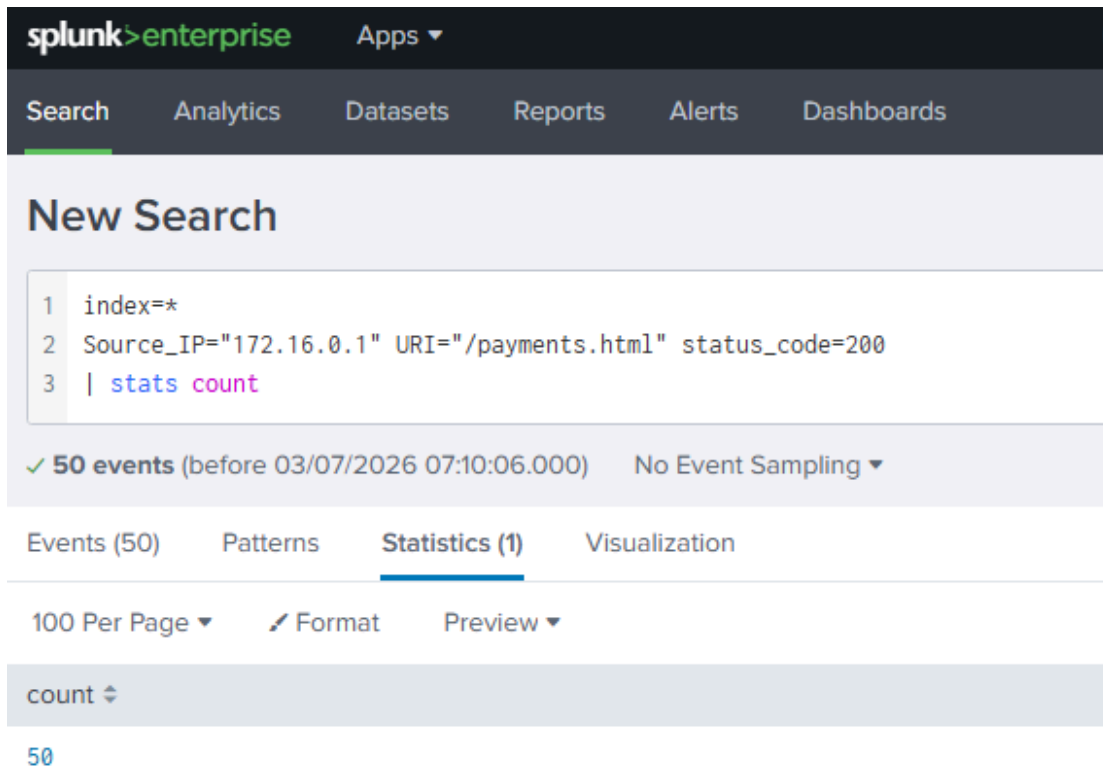
URI ⇅

- /pictures.html
- /contact.html
- /trainings.html
- /about.html
- /index.html
- /restricted.html
- /payments.html

Add another statistics table to your dashboard to view the Source\_IP, URI, and status\_code fields. How many times did 172.16.0.1 receive the status\_code 200 from /payments.html?

50





This is the hint you're looking for: Try this query: `index = web_logs | stats count by Source_IP URI status_code | sort - count`

## Task 5 - Extending Splunk's Functionality

**Splunk Enterprise Security:** Provides a complete SOC framework with security visibility, correlation searches, threat intelligence, risk scoring, MITRE ATT&CK mapping, and investigation workflows.

**Splunk UEBA:** Detects abnormal user and entity behavior by analyzing activity over time and assigning risk scores to identify insider threats, compromised accounts, and suspicious activity.

**Splunk SOAR:** Automates incident response using playbooks that perform actions such as IP reputation checks, host isolation, and account disabling, reducing response time and repetitive manual work.

**Answer the questions below**

Which feature in Splunk Enterprise Security provides visibility into SOC efficiency and performance?

SOC Operations

What does Splunk SOAR use to automate response actions based on conditions, filters, and decision logic?

Playbooks

## **Task 6 - Conclusion**

Splunk reports and dashboards improve event visibility, while detection rules, ES, and SOAR enable threat detection, correlation, automation, and incident response—turning Splunk into a complete SOC platform.